Q1. DIRECTIONS *for question 1:* Select the correct alternative from the given choices.
A right-angled triangle is cut along a line parallel to the hypotenuse of the triangle, such that the length of the hypotenuse of the smaller triangle is 35% less than that of the initial triangle. If the area of the smaller triangle is x percent of the area of the initial triangle, find the value of x, rounded off to the nearest whole number.

- a) **42**
- b) **35**
- c) **12**
- d) **65**

Q2. DIRECTIONS *for question 2:* Type in your answer in the input box provided below the question.

If $f(n) = 1(1!) + 2(2!) + 3(3!) + \ldots\ldots + n(n!)$, find the remainder when $f(117) + f(111)$ is divided by 100.

Q3. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.
What is the minimum number of integers from 7 to 77 (both inclusive) that must be chosen so that at least one multiple of 3 is included?

- a) **3**
- b) **58**
- c) **59**
- d) **49**

Q4. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

If $|x - 3|.(y^2 - 4x + 4) < 0$, which of the following is definitely true?

- a) $-2 < x < 3$
- b) $y > 2$
- c) $-2 < y < 1$
- d) $x > 1$

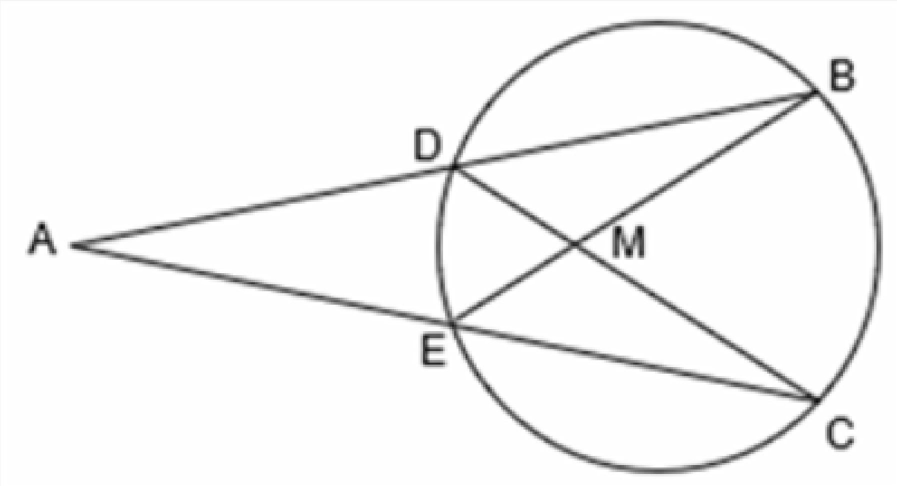
Q5. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

Two vessels, A and B, contain a mixture of kerosene and petrol in the ratio 4 : 3 and 8 : 11 respectively. If the contents of A and B are mixed, which of the following cannot be the ratio of kerosene and petrol in the mixture thus formed?

- a) **13 : 17**
- b) **11 : 9**
- c) **10 : 7**
- d) **7 : 9**

Q6. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

In the figure below, if DM = 4 cm, MC = 18 cm and $\frac{AE}{AD} = \frac{9}{11}$, then find the measure (in cm) of MB.



a) **6**

b) **8**

c) **10**

d) **12**

Q7. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.
Consider the following three equations:

$2x + 3y + 4z = 33$

$4x + 2y + 3z = 29$

$3x + 4y + 2z = 28$

Which of the following equations is inconsistent with the above equations?

- a) $\mathbf{x + y + z = 10}$
- b) $\mathbf{6x + 5y + 7z = 62}$
- c) $\mathbf{y + z - 2x = 4}$
- d) $\mathbf{5x + 3y + z = 23}$

Q8. DIRECTIONS *for questions 3 to 9:* Select the correct alternative from the given choices.

If $x \neq 0$, what is the product of the roots of the equation $\frac{(4x^{4001} + 5x^{4000})}{(6x^{3999})} = 4$?

a) **4**

b) **–24**

c) **12**

d) **–6**

Q9. **DIRECTIONS** *for questions 3 to 9:* Select the correct alternative from the given choices.

Tarun and Varun invested in a business in the ratio of 4 : 5, at the beginning of a year. If Tarun withdrew from the partnership after eight months, what percentage of the annual profit will Tarun receive?

a) **34.78%** ✓ **Your answer is correct**

b) **33.33%**

c) **29.63%**

d) **35%**

Q10. DIRECTIONS *for question 10:* Type in your answer in the input box provided below the question.

A transport agency has five carriers, each of capacity ten tonnes. The carriers are scheduled such that the first carrier makes a trip every day, the second carrier makes a trip every second day, the third makes a trip every third day and so on. Find the maximum number of times in a year that it is possible to dispatch a total shipment of 50 tonnes in a single day. The operations start on the 7th of January and continue till the end of the year, i.e., the 31st of December, without any holidays.

Q11. DIRECTIONS *for questions 11 to 13:* Select the correct alternative from the given choices.

A country wanted to select four mixed-doubles tennis teams for the upcoming Olympics from 6 female players and 6 male players available for selection. If the selection panel chose the players at random, then in how many ways they could have done that? (A mixed-doubles tennis team comprises one male and one female player)

- a) **129600**
- b) **64800**
- c) **5400**
- d) **1800**

Q12. DIRECTIONS *for questions 11 to 13:* Select the correct alternative from the given choices.

If the equations of the two tangents drawn to a circle C from a point P are $4x + 3y - 7 = 0$ and $3x - 4y + 1 = 0$, then the centre of the circle C will definitely lie on the pair of straight lines represented by

- a) $(x + 6y - 7)(6x + y - 7) = 0.$
- b) $(x + 7y - 8)(7x - y - 6) = 0.$
- c) $(x - 6y + 5)(6x + y - 7) = 0.$
- d) $(x - 7y + 6)(7x + y - 8) = 0.$

Q13. **DIRECTIONS** *for questions 11 to 13:* Select the correct alternative from the given choices.
The number of bacteria in a laboratory doubles at the end of every 20 minutes. If the number of bacteria is 364 at 8 a.m., which of the following is the earliest time at which the number of bacteria will be 10,000 or more?

- a) **9:00 a.m.**
- b) **9:20 a.m.**
- c) **9:40 a.m.** ✓ **Your answer is correct**
- d) **10:00 a.m.**

Q14. **DIRECTIONS** *for questions 14 and 15:* Type in your answer in the input box provided below the question.

In Hitami theatre, the price of a ticket, which is paid for just before exiting the theatre, varies as $P = Ax + By + C$, where P is the price of the ticket, x is the number of hours spent inside the theatre and y is the number of popcorns purchased, with A, B and C being constants. Today, Ram spent three hours inside the theatre and purchased seven popcorns. He would have spent ₹300 less on the ticket had he either spent 2.5 hours less and bought three popcorns more OR spent 1.5 hours more and purchased five popcorns less.

What is the maximum number of hours that Ram could have spent inside the theatre with the amount that he spent?

Q15. **DIRECTIONS** *for questions 14 and 15:* Type in your answer in the input box provided below the question.

The difference between the simple interest and the compound interest accrued in two years on a certain sum at a certain rate of interest is ₹80. If the simple interest for the second year is ₹1000, find the sum (in ₹).

Q16. DIRECTIONS *for question 16:* Select the correct alternative from the given choices.

The pair of straight lines $y^2 = 2x^2$ meet the graph of $y^2 = 4x$ at the origin and also at points P and P', forming two closed loops. The ratio of the area of the upper loop to that of the lower loop is

- a) **1 : 2.**
- b) **1 : 1.**
- c) **4 : 1.**
- d) **2 : 1.**

Q17. DIRECTIONS *for question 17:* Type in your answer in the input box provided below the question.

If $1 + 3 + 5 + \ldots + (2n - 1) = 1 + m(m+1)(m+2)(m+3)$, where m and n are natural numbers, find the value of n, when $m = 25$.

Q18. DIRECTIONS *for question 18:* Select the correct alternative from the given choices.

A student was asked to first divide a number by 7, then add 6 to the result and write down the sum as the answer. He, however, first added 6 to the number and then divided the sum by 7, getting 25 as the result, which he wrote down as the answer. Which of the following should have been the correct answer?

- a) $\frac{211}{7}$
- b) $\frac{169}{7}$
- c) $\frac{127}{7}$
- d) $\frac{169}{6}$

Q19. **DIRECTIONS** *for question 19:* Type in your answer in the input box provided below the question.

The total age of a few eight-year-old children and a few seven-year-old children is 70 years. If there is at least one eight-year-old and at least one seven-year-old, in how many ways can a team be selected from these children, such that the sum of the ages of the children in the team is 54?

Q20. **DIRECTIONS** *for questions 20 and 21:* Select the correct alternative from the given choices.
Given below are two statements I and II:

I. The ratio of the altitudes of a triangle ABC is 2 : 5 : 6.

II. The ratio of the altitudes of a triangle DEF is 3 : 4 : 8.

Which of the above two statements can be true?

- a) **Only I**
- b) **Only II**
- c) **Both I and II**
- d) **Neither I nor II**

Q21. **DIRECTIONS** *for questions 20 and 21:* Select the correct alternative from the given choices.
Maaldar Reddy bequeathed his property comprising ‘A’ acres of land to his three sons, such that the areas of the shares of land given to the sons were in geometric progression. If the maximum difference between the shares of any two sons is 385 acres and the least possible sum of the shares of any two sons is 770 acres, then find the value of A.

- a) **650**
- b) **975**
- c) **1463**
- d) **1580**

Q22. DIRECTIONS *for question 22:* Type in your answer in the input box provided below the question.
If P is 30% more efficient than Q and can complete a certain job in 46 days, how many days would P and Q together take to complete the same job?